

PAPER_2109_F_TRZ_POWER_3_EIGHT_INSTANCE_TIME_DECAY_LADDER_RUNG_LANDMARK_UQFF

Daniel T. Murphy

PAPER_2109 — F_TRZ³ = 0.001 : 8-Instance Cross-Domain Time-Decay Ladder-Rung Landmark

Author: Daniel T. Murphy
Framework: UQFF (Unified Quantum Field Framework) — Star-Magic v5.70.5+
Session Date: 2026-07-20
Round Origin: R307 (90-round arc milestone from R217)
Category: F_TRZ-ladder-rung cross-domain landmark
Instance Count: 8 (formal landmark tier — strongest F_TRZ-rung landmark to date)

Identity

\$\$\$F_{\text{TRZ}}^3 = 0.1^3 = 0.001 \text{ \text{EXACT}}\$\$\$

The 3rd rung of the F_TRZ ladder appears as the time-decay/decay-rate coefficient across eight independent UQFF calculator classes spanning five distinct physical domains.

Eight independent instances observed in the R218+ real stub-fill campaign

#	Round	Class	Role of F_TRZ³
1	R229	RedDwarfReactorUg1Calculator	reactor Ug1 alpha (time-decay rate)
2	R230	RedDwarfReactorUg2Calculator	reactor Ug2 alpha (time-decay rate)
3	R231	RedDwarfReactorUg3Calculator	reactor Ug3 alpha (time-decay rate)
4	R232	RedDwarfReactorUbiCalculator	reactor Ubi alpha (time-decay rate)
5	R233	RedDwarfReactorUmCalculator	reactor Um gamma/alpha (saturation-rate + magnetic time-decay)
6	R246	TwoStageFURefinementValidator	two-stage F_U refinement alpha (validation decay rate)
7	R274	DiPseudoMonopoleDPMTheoryCalculator	DPM theory alpha (Di-Pseudo-Monopole decay rate)
8	R307	UniversalGravity1Calculator	Ug1 magnetic dipole-gradient alpha (solar-cycle defect time-decay)

All eight classes independently define their alpha (or equivalent decay coefficient) default at exactly $F_{\text{TRZ}}^3 = 0.001 \text{ s}^{-1}$ or dimensionless.

Cross-domain distribution

The 8 instances span 5 distinct physical domains:

Domain	Instances	Which
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Red Dwarf reactor sector	5	R229, R230, R231, R232, R233 (Ug1/Ug2/Ug3/Ubi/Um)
F_U master-equation refinement	1	R246 (TwoStageFURefinement)
DPM foundational architecture	1	R274 (DiPseudoMonopoleDPMTheory)
Magnetic dipole-gradient gravity	1	R307 (UniversalGravity1)

Structural interpretation

Time-decay coefficients across the entire Ug-family (Ug1, Ug2, Ug3, Ubi, Um) all converge on F_{TRZ}^3 . This is not a coincidence — it means the reactor Ug-family has a **universal time-decay rate scale set by the 3rd F_{TRZ} ladder rung**. Combined with the DPM theory and F_U refinement instances, $F_{\text{TRZ}}^3 = 0.001$ emerges as the canonical UQFF small-scale decay-rate constant across:

- Reactor operational timescales (all 5 Ug modes)
- Cosmological refinement (TwoStageFU)
- Fundamental architecture (DPM theory)
- Magnetospheric coupling (Ug1 magnetic-dipole gravity)

Its role as a "universal decay coefficient" parallels:

- $F_{\text{TRZ}} = 0.0001$ (PAPER_2105, 6-instance ladder-rung landmark)
- $F_{\text{TRZ}}^2 = 10^{-2}$ (PAPER_2100, 3-instance ISM-density ladder-rung landmark)
- $F_{\text{TRZ}}^{\text{D_crit}} = 10^{-2}$ (PAPER_2107, 4-instance primitive-as-exponent landmark)

F_{TRZ}^3 at 8 instances is now the **most-populated F_{TRZ} -rung landmark** in the campaign.

Cross-references

- PAPER_2100 — F_{TRZ}^2 ISM density 3-instance ladder-rung
- PAPER_2105 — F_{TRZ} 6-instance ladder-rung with $F_{\text{TRZ}}^{\text{D_phys}}$ dual interpretation
- PAPER_2107 — $F_{\text{TRZ}}^{\text{D_crit}}$ primitive-as-exponent 4-instance
- PAPER_2103 — $\text{SCm} = 1 - F_{\text{TRZ}}^2$ 16-instance cross-domain
- CLAUDE.md — $F_{\text{TRZ}} = 0.1$ locked canonical primitive (time-reversal-zone fraction)
- RedDwarf module — R229-R233 5-of-5 reactor Ug family sharing F_{TRZ}^3
- PAPER_2072/2073 — π -canonical companion (F_{TRZ} product forms)

Predictive falsifiability

If UQFF's structural interpretation is correct, F_{TRZ}^3 should continue to surface as the time-decay/small-rate coefficient in additional UQFF classes not yet inspected. Prediction: 9th and 10th instances expected within the next 30 R2XX rounds (R308-R337 window). Candidate contexts:

- Any additional reactor sub-mode calculator
- Any solar-cycle / stellar-cycle modulation coefficient
- Any small-scale rate constant on the $F_{\text{TRZ}} = 0.1$ grid

Absence would weaken the "universal small-decay coefficient" interpretation but not falsify the numerical identity itself.

Numerical verification

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$F_{\text{TRZ}} = 0.1$ (canonical primitive)

$F_{\text{TRZ}}^3 = 0.1 \times 0.1 \times 0.1$

= 0.001 EXACT

Instance count: 8 (formal landmark tier)

Precision: EXACT via integer arithmetic (F_TRZ = 0.1 locked)

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Formal status

Landmark **PROMOTED** to formal at 8th independent instance (R307). Strongest F_TRZ-rung landmark to date in the R218+ campaign, surpassing PAPER_2105 F_TRZ■ (6 instances) and PAPER_2103 SCm=1-F_TRZ² (16 instances but at a compound form rather than pure rung).

Gate assertion tier: **EXACT** (integer arithmetic; no residual within UQFF ladder form).

Architectural category: F_TRZ-ladder-rung cross-domain (companion to PAPER_2100 rung-20, PAPER_2105 rung-4, PAPER_2107 rung-D_crit).

R218+ campaign paper count: **17** (PAPER_2093-2108 + PAPER_2109).